PillCare Reminder V01

Complete 10-Module Enterprise Documentation Architecture
Product, Technical Architecture, Security & Operations Master Specification

Module: 01 Product

Document: PRD.md

PillCare - Product Requirements Document (PRD)

1. Executive Summary

PillCare is a healthcare technology platform engineered to maximize patient medication adherence, reduce accidental missed doses, and empower caregivers through automated emergency escalation. Powered by a hybrid offline-first architecture, PillCare ensures uninterrupted local alarm notifications even without internet connectivity while providing real-time telemetry to caregivers and medical providers.

2. Target Personas

1. **Senior Patients (Chronic Illness)**: Requires high-contrast UI, large touch targets ($\ge 48\text{dp}$), voice alarms, and automatic dose logging.

2. **Busy Professionals**: Requires rapid optical prescription scanning (OCR), 1-tap confirmation, calendar integration, and refill automation.

3. **Caregivers & Guardians**: Requires real-time SMS/WhatsApp missed dose alerts, refill status visibility, and multi-patient management.

4. **Enterprise Healthcare Providers**: Requires adherence risk heatmaps, revenue analytics, and custom feature control via Mission Control.

3. Core Product Capabilities & Feature Specifications

3.1 Medication Scheduling & Smart Alarms

- Flexible schedules: Daily, specific days, alternate days, interval cycles, and AS NEEDED (PRN).
- Exact alarm scheduling using native Android `AlarmManager` / iOS `UserNotifications` bypassing OS battery optimization (`USE_EXACT_ALARM`).
- Food relationship tagging: `before_food`, `with_food`, `after_food`, `empty_stomach`.

3.2 Caregiver Shield (Emergency Escalation)

- Multi-tier escalation: Primary caregiver, secondary guardian, emergency contact.
- Automated WhatsApp & SMS alert dispatch upon missed medication window ($\ge 15\text{ mins}$).
- Caregiver Web Portal link dispatched via secure single-use token.

3.3 Swiggy / Zomato Flash Announcement Engine

- Dynamic top banner announcements pushed in real-time from Admin Control Center.
- Audience targeting: Free plan users, Pro subscribers, or Enterprise hospital groups.
- Live conversion tracking (CTR & impression logging).

3.4 Refill Radar & Automated Inventory

- Real-time pill inventory decrement upon dose confirmation.
- Refill threshold alerts (e.g., alert when stock drops below 5 pills).
- 1-Tap pharmacy reordering via partner API / WhatsApp message templates.

3.5 AI Yoga & Wellness Therapy

- Custom pranayama and gentle yoga video routines tailored to specific medical conditions (Hypertension, Diabetes, Joint Stiffness).
- Daily compliance journal logging and streak tracking.

4. Monetization & Subscription Tiers

Feature	Free Tier	Pro Moksha (■99/mo)	Enterprise Hospital
Active Medications	Up to 3	Unlimited (999)	Unlimited
Caregiver Alerts	1 Contact	Unlimited Caregivers	Dedicated Escort
OCR Scanner	2 Scans/mo	Unlimited Scans	EHR Auto-Sync
Yoga Therapy	Basics	Full HD Library	Custom Routines
Feature Overrides	Default	Admin Modifiable	Custom Hospital SLA

5. Non-Functional Requirements & Constraints

- **Offline-First**: All alarms, schedules, and dose logs operate 100% offline via SQLite (`AsyncStorage`).
- **Telemetry Latency**: Event dispatch latency $<100\text{ms}$ when connected.
- **Privacy & Compliance**: Zero sale of personal health information; HIPAA & DISHA compliant data boundaries.

Document: Product_Roadmap.md

PillCare - Product Roadmap (2026 - 2027)

1. Vision Statement

To build India's most trusted offline-first medication adherence and preventive health intelligence platform, bridging the gap between patients, family caregivers, and clinical care teams.

2. Strategic Horizon & Milestones

```
+-------------------------------------------------------------------------------+
| Q3 2026: V1.0 Foundation & Core Experience                                    |
|  - Offline-first medication alarms & snooze loops                            |
|  - Caregiver Shield WhatsApp emergency escalation                           |
|  - Admin Mission Control & Swiggy/Zomato Flash Announcements                 |
```

```
|  - Standalone Android APK & EAS Cloud Pipelines                                |
+--------------------------------------------------------------------------------+
                                     |
                                     v
+--------------------------------------------------------------------------------+
| Q4 2026: V1.1 Healthcare Intelligence & Monetization Expansion                 |
|  - JP Morgan Financial Dashboard & real-time MRR/ARR analytics                 |
|  - Adherence Risk Radar & Condition Heatmaps                                   |
|  - AI Yoga Therapy Compliance Tracking                                         |
|  - Google Sign-In & Multi-device Cloud Backup                                  |
+--------------------------------------------------------------------------------+
                                     |
                                     v
+--------------------------------------------------------------------------------+
| Q1 2027: V2.0 Enterprise Clinical & EHR Integration                            |
|  - Hospital EHR / FHIR API Data Connectors                                     |
|  - Smart Pillbox Bluetooth (BLE) Syncing                                       |
|  - AI Voice Prescription Parser & Drug-Drug Interaction Warnings               |
|  - Regional Language Audio Alarms (Hindi, Tamil, Telugu, Kannada, Bengali)     |
+--------------------------------------------------------------------------------+
```

3. Feature Breakdown & Timeline

Q3 2026 (Completed)

- [x] High-precision local notifications & exact alarm scheduling.
- [x] Caregiver Alert escalation via WhatsApp template engine.
- [x] Admin Control Center (`admin.pillcare.in`) with 2-way manual overrides.
- [x] Refill Radar inventory automation.

Q4 2026 (Active)

- [x] Healthcare Intelligence Radar (Adherence, Refill GMV, Risk Heatmap).
- [x] JP Morgan Financial KPIs (MRR, ARR, ARPU, LTV, Churn).
- [ ] iOS HealthKit Integration & WatchOS Companion App.
- [ ] Direct WhatsApp Bot for 1-click dose confirmation.

Q1 2027 (Future Exploration)

- [ ] Hospital OPD Patient Portal integration.
- [ ] BLE Smart Pill Bottle hardware synchronization.
- [ ] Pharmacovigilance adverse event reporting API.

Document: Feature_Requirements.md

PillCare - Feature Requirements Matrix

1. Functional Requirements Matrix

ID	Feature Name	Priority	Target User	Description	Acceptance Criteria

FR-01	Medication Creation	**P0**	Patient	Create medication schedule with custom dosages, times, food relation, and stock limits.	Must save locally instantly and schedule exact Android/iOS local alarm.
FR-02	Alarm & Snooze	**P0**	Patient	Trigger full-screen high-priority alert at scheduled time with 5-min, 10-min, 15-min snooze options.	Must trigger sound/vibration even in Do Not Disturb / Low Power Mode.
FR-03	Caregiver Shield	**P0**	Caregiver	Dispatch WhatsApp alert to registered caregiver if medication is not marked taken within 15 minutes.	Alert logged in database and dispatched via WhatsApp API payload.
FR-04	OCR Scanner	**P1**	Patient	Scan physical prescription using camera to extract drug name, dosage, and frequency.	Text recognized with $>90\%$ accuracy and auto-populates creation form.
FR-05	Refill Radar	**P1**	Patient	Monitor remaining pill count and alert user when stock drops below user-defined threshold.	Stock decrements on dose confirmation; refill modal pops up automatically.
FR-06	AI Yoga Routine	**P2**	Patient	Provide curated video flows for blood pressure, glucose, and anxiety management.	Session logs completion date, duration, and updates patient wellness journal.
FR-07	Flash Banners	**P1**	Admin / User	Display real-time p romo/announceme nt banners on the Today tab.	Pushed from Admin Panel; respects plan targeting (Free vs Pro).
FR-08	2-Way Overrides	**P0**	Admin	Manually override user plan or specific feature flags (e.g. enable OCR scanner for Free user).	Precedence rule: Admin override > Plan default rule.

2. Dependencies & Assumptions

- Native Android `SCHEDULE_EXACT_ALARM` and `USE_EXACT_ALARM` permissions must be granted by OS.
- Caregiver alert dispatch requires active network connectivity on device or server-side cron trigger.

- Offline logs synchronize seamlessly once device reconnects to internet.

Module: 02 UI UX

Document: User_Flows.md

PillCare - User Flows & Patient Journeys

1. Primary Patient Journey (Dose Logging & Caregiver Shield)

```
sequenceDiagram
    autonumber
    actor Patient
    participant App as Mobile App (Expo)
    participant LocalDB as SQLite (Local)
    participant Notification as OS Alarm Manager
    participant Backend as FastAPI Backend
    participant Caregiver as Caregiver (WhatsApp)

    Patient->>App: Opens App & Schedules Medication
    App->>LocalDB: Saves Medication & Initial Stock (e.g. 30 pills)
    App->>Notification: Schedules Exact Local Alarm (e.g. 08:00 AM)

    Note over App,Notification: Time elapses to 08:00 AM
    Notification->>Patient: Fires Full-Screen Audio Alarm & High-Priority Banner

    alt Patient Takes Medication Promptly
        Patient->>App: Taps "MARK AS TAKEN"
        App->>LocalDB: Log Taken State & Decrement Stock (29 pills remaining)
        App->>Notification: Cancel Missed Dose Grace Timer
        App->>Backend: Sync Log Telemetry (Adherence +1)
    else Patient Misses Medication (15-Minute Window Expires)
        Note over App,Backend: Grace Period Expires (15 mins)
        App->>Backend: Dispatch Missed Dose Event
        Backend->>Caregiver: Send WhatsApp Alert ("Patient missed Metformin at 08:00 AM")
        Caregiver->>Patient: Calls / Reminds Patient directly
    end
```

2. Refill Automation Flow

```
flowchart TD
    A[Patient Takes Medication] --> B[Local Stock Decremented]
    B --> C{Remaining Stock <= Refill Threshold?}
    C -- No --> D[Normal State Retained]
    C -- Yes --> E[Trigger Refill Alert Modal]
    E --> F{Patient Choice}
    F -- 1-Tap Refill --> G[Update Stock + Add Pack Count]
    F -- Order Pharmacy --> H[Launch Pharmacy WhatsApp / Partner Link]
    G --> I[Refill Radar Updated in Database]
```

3. Admin Overrides & Flash Announcement Flow

```
sequenceDiagram
    actor Admin
    participant Dashboard as admin.pillcare.in
    participant Backend as FastAPI Backend
    participant App as Mobile App
```

```
    Admin->>Dashboard: Publish Swiggy/Zomato Flash Offer (50% Off Moksha Pro)
    Dashboard->>Backend: POST /admin/announcements
    Backend-->>Dashboard: 200 OK (Announcement ID: ann_123)

    App->>Backend: GET /api/announcements/active?plan=free
    Backend-->>App: Return Active Flash Offer Payload
    App->>App: Render Dynamic Flash Banner at top of Today Screen
```

Document: Wireframes.md

PillCare - Wireframes & Screen Layout Specifications

1. Main Navigation Hierarchy (Tab Navigator)

```
+-------------------------------------------------------------------------+
|  [Today]   |   [Medications]   |   [Journal]   |   [Caregiver]       |
+-------------------------------------------------------------------------+
```

2. Today Screen Wireframe

```
+-------------------------------------------------------------------------+
|  PILLCARE                                          [Profile / Upgrade]|
|  Tuesday, 15 September 2026                                            |
+-------------------------------------------------------------------------+
|  ■ FLASH ANNOUNCEMENT BANNER (Swiggy/Zomato Flash Offer)               |
|  "Special 50% Off Annual Moksha Upgrade - Claim Now!"                |
+-------------------------------------------------------------------------+
|  DAILY ADHERENCE PROGRESS                                              |
|  [==========================>        ] 75% Completed (3/4 Meds)       |
+-------------------------------------------------------------------------+
|  UPCOMING MEDICATION                                                   |
|  +-------------------------------------------------------------+  |
|  |  [Icon] Metformin 500mg                                      |  |
|  |         Time: 08:00 AM  |  Relation: With Food                |  |
|  |         Stock: 12 pills remaining                            |  |
|  |  [ MARK AS TAKEN ]          [ SNOOZE 15M ]                   |  |
|  +-------------------------------------------------------------+  |
+-------------------------------------------------------------------------+
|  DAILY WELLNESS YOGA                                                   |
|  +-------------------------------------------------------------+  |
|  |  [Play] Anulom Vilom Pranayama (5 mins)                      |  |
|  |         Condition Target: Hypertension / Stress Reduction    |  |
|  |  [ START SESSION ]                                           |  |
|  +-------------------------------------------------------------+  |
+-------------------------------------------------------------------------+
```

3. Medications Management Screen Wireframe

```
+-------------------------------------------------------------------------+
|  MY MEDICATIONS                                    [ + Add Med ]   |
+-------------------------------------------------------------------------+
|  [ Search medications... ]                                             |
+-------------------------------------------------------------------------+
|  ACTIVE MEDICATION LIST                                                |
|  1. Amlodipine 5mg                                                     |
```

```
|     Schedule: Daily at 09:00 AM                                        |
|     Stock: 24 pills  | Refill Threshold: 5                             |
|                                                                        |
|  2. Metformin 500mg                                                    |
|     Schedule: Twice Daily (08:00 AM, 08:00 PM)                         |
|     Stock: 3 pills  ■■ LOW STOCK REFILL ALERT                           |
|     [ REFILL NOW (+30) ]                                               |
+------------------------------------------------------------------------+
```

4. Caregiver Shield Screen Wireframe

```
+------------------------------------------------------------------------+
|  CAREGIVER SHIELD                                  [ Status: ACTIVE ] |
+------------------------------------------------------------------------+
|  EMERGENCY ESCALATION CONTACTS                                         |
|  Primary Caregiver: Ramesh Kumar (+91 9876543210)                      |
|  Channel: WhatsApp Escalation Enabled                                  |
+------------------------------------------------------------------------+
|  RECENT ESCALATION ALERTS DISPATCHED                                   |
|  - 14 Sep, 08:15 PM: Metformin 500mg missed -> WhatsApp Sent           |
|  - 12 Sep, 08:15 AM: Amlodipine 5mg missed -> WhatsApp Sent            |
+------------------------------------------------------------------------+
```

Document: UI_Specification.md

PillCare - UI Specification & Accessibility Guidelines

1. Accessibility First Design Principles (Senior-Friendly Standards)

1. **Minimum Touch Targets**: All interactive buttons, cards, and checkboxes must strictly satisfy $\ge 48\text{dp} \times 48\text{dp}$ touch area.

2. **High-Contrast Text**: Text elements meet WCAG AAA contrast ratio ($\ge 7:1$) against background surfaces.

3. **Typography Scale**: Default body text $\ge 16\text{sp}$; headline text $\ge 24\text{sp}$.

4. **Haptic & Audio Feedback**: Tactile haptic pulse (`expo-haptics`) paired with custom sound effects (`expo-notifications`) for dose confirmation.

2. Color System Tokens

```
:root {
  /* Brand Core Colors */
  --color-primary-dark-teal: #002B2B;  /* Deep Emerald Background */
  --color-primary-emerald:   #0A5C36;  /* Primary Brand Buttons */
  --color-accent-lavender:   #9184D9;  /* Wellness Accent */
  --color-warning-amber:     #D97706;  /* Low Stock Alerts */
  --color-danger-crimson:    #DC2626;  /* Missed Dose / Critical */

  /* Surface Tokens (Dark Glassmorphism) */
  --color-surface-card:      rgba(10, 40, 40, 0.75);
  --color-surface-border:    rgba(255, 255, 255, 0.12);
  --color-text-primary:      #FFFFFF;
  --color-text-secondary:    #9CA3AF;
}
```

3. Responsive Breakpoints & Device Layout Rules

- **Mobile Handsets (<600dp width)**: Single-column vertical flow with sticky bottom action tabs.
- **Tablets (>=600dp width)**: Master-detail split view (Medication list on left, detailed scheduling/log history on right).

Document: Design_System.md

PillCare - Stitch Design System Guidelines

1. System Overview

The **PillCare Stitch Design System** combines modern dark glassmorphism, fluid micro-interactions, and medical clarity to evoke trust, calm, and precision.

2. Component Specifications

2.1 Medication Card Component (`MedicationCard.tsx`)

- **Background**: Glassmorphic blur (`expo-blur` intensity 40) with `#0A2828` tinted container.
- **Border**: `1px solid rgba(255, 255, 255, 0.12)` with `16px` border-radius.
- **Elevation**: Subtle drop shadow (`shadowColor: "#000", shadowOpacity: 0.3, shadowRadius: 8`).

2.2 Primary CTA Button

- **Height**: `52px` minimum.
- **Background**: Linear Gradient (`#0A5C36` $\rightarrow$ `#004D25`).
- **Typography**: Semibold 16sp white uppercase text with 1px letter spacing.
- **Active State**: Scale down animation (`transform: [{ scale: 0.96 }]`) paired with medium haptic pulse.

2.3 Flash Announcement Banner Component (`FlashAnnouncementBanner.tsx`)

- **Gradient Fill**: Swiggy/Zomato brand gradient (`#FF5200` $\rightarrow$ `#E23744` or Deep Purple/Gold for Pro upgrades).
- **Positioning**: Fixed top banner above Today screen adherence progress card.
- **Dismissability**: Includes dismiss cross icon with 24-hour cookie suppression.

Module: 03 Technical

Document: SRS.md

PillCare - System Requirements Specification (SRS)

1. Introduction

This System Requirements Specification (SRS) defines the software, hardware, operational, and non-functional requirements for the PillCare Medication Adherence & Healthcare Intelligence system, following IEEE 830 standards.

2. System Performance SLAs & Specifications

2.1 Latency SLA

- **Local Medication Alarm Trigger**: 0ms delay (Fired locally by native OS `AlarmManager` / `UserNotifications`).
- **API Response Latency**: $\le 100\text{ms}$ for 99th percentile of REST requests under normal network conditions.
- **Caregiver Alert Escalation**: Dispatch initiated within $\le 30\text{ seconds}$ of grace period expiration.

2.2 System Throughput & Scale

- **Backend Concurrency**: Cloud Run container scaling to handle $\ge 10,000$ requests/minute.
- **Database I/O**: Local SQLite transactions execute in $<5\text{ms}$.

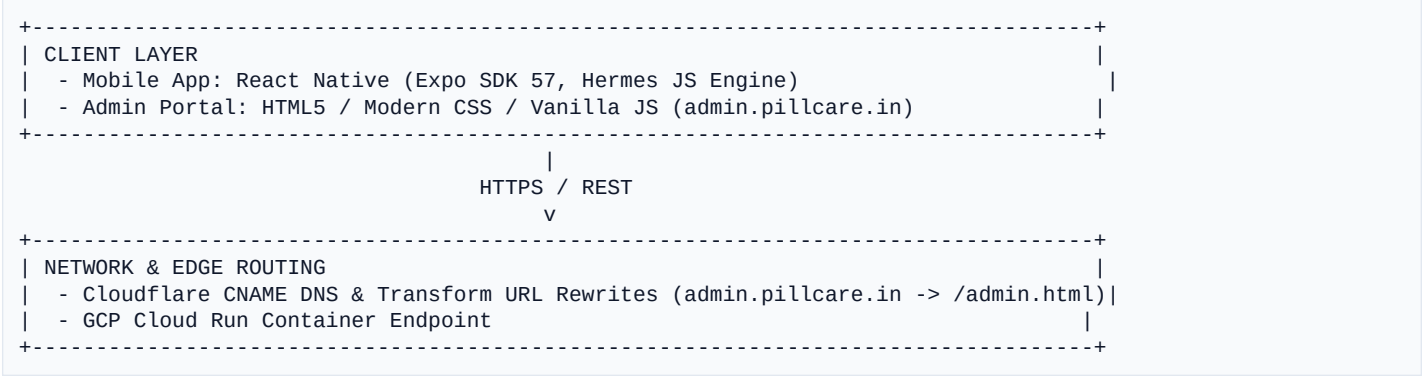
3. Hardware & Platform Compatibility Matrix

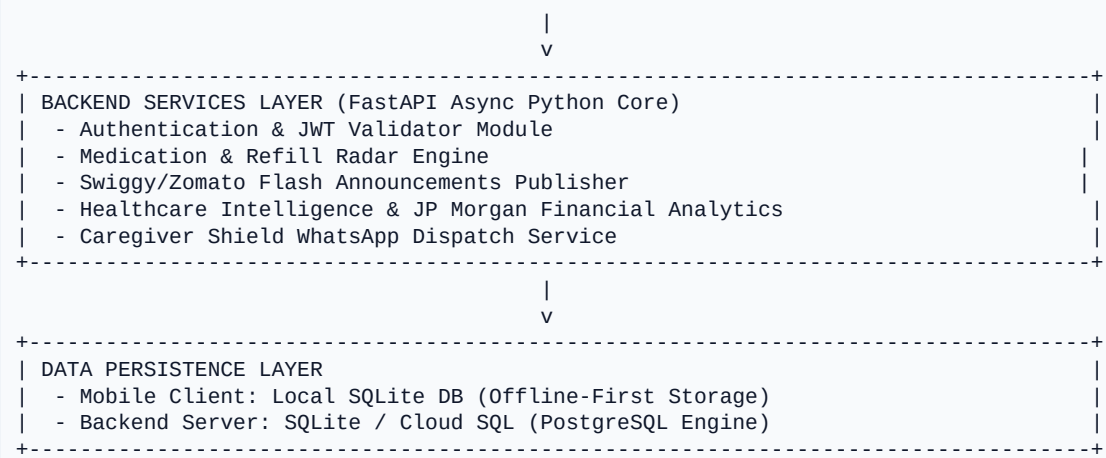
- **Android**: Android 8.0 (API Level 26) up to Android 15/16 (API Level 36).
- **iOS**: iOS 15.0 and above.
- **Web Admin**: Chrome 100+, Safari 15+, Edge 100+ (Responsive $\ge 1024\text{px}$).

Document: System_Architecture.md

PillCare - System Architecture Specification

1. High-Level Architecture Diagram





2. Key Subsystem Components

1. **Local Notification Engine (`expo-notifications`)**: Manages native alarms bypasses battery optimization (`SCHEDULE_EXACT_ALARM`).

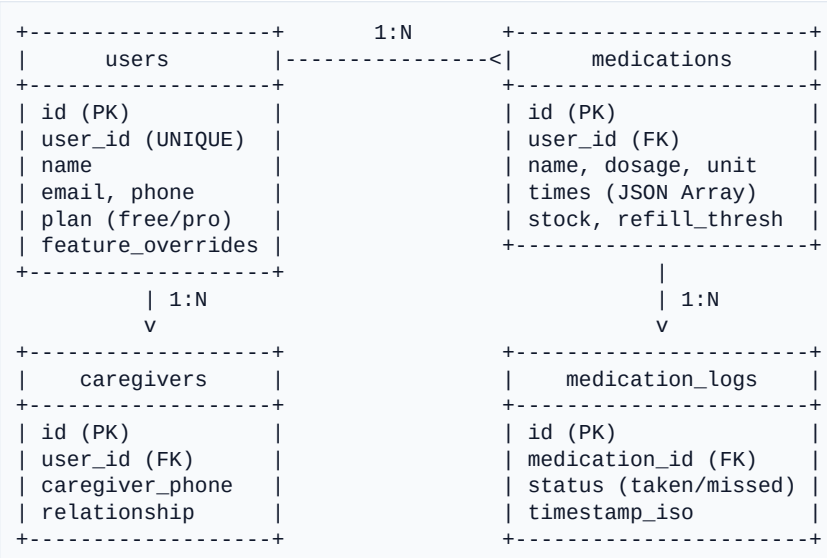
2. **FastAPI Backend (`backend/server.py`)**: Asynchronous Python ASGI microservice hosting REST endpoints, analytics calculators, and admin security gatekeeper.

3. **Caregiver Alert Escalation Service**: Listens for unconfirmed dose timestamps and dispatches WhatsApp template alerts to registered caregiver numbers.

Document: Database_Design.md

PillCare - Database Design & ERD Specification

1. Relational Entity Relationship Diagram (ERD)



2. DDL Schema Definitions (SQLite / PostgreSQL)

2.1 `users` Table

```
CREATE TABLE users (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    user_id TEXT UNIQUE NOT NULL,
    email TEXT,
    name TEXT,
    plan TEXT DEFAULT 'free',
    feature_overrides TEXT DEFAULT '{}',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

2.2 `medications` Table

```
CREATE TABLE medications (
    id TEXT PRIMARY KEY,
    user_id TEXT NOT NULL,
    name TEXT NOT NULL,
    generic_name TEXT,
    dosage TEXT NOT NULL,
    unit TEXT NOT NULL,
    times TEXT NOT NULL, -- JSON Array: ["08:00", "20:00"]
    start_date TEXT NOT NULL,
    stock INTEGER DEFAULT 30,
    refill_threshold INTEGER DEFAULT 5,
    food_relation TEXT DEFAULT 'with_food',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY(user_id) REFERENCES users(user_id)
);
```

2.3 `announcements` Table

```
CREATE TABLE announcements (
    id TEXT PRIMARY KEY,
    title TEXT NOT NULL,
    body TEXT NOT NULL,
    type TEXT DEFAULT 'offer',
    target_plan TEXT DEFAULT 'all',
    link_url TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Document: API_Documentation.md

PillCare - REST API Specification

1. Authentication Headers

All protected client endpoints require Bearer JWT token:

```
Authorization: Bearer <session_token>
```

All protected admin endpoints require Admin Secret Key:

```
x-admin-secret: pillcare_admin_secret_key_2026
```

2. Core API Endpoints Reference

2.1 User & Feature Management

• `GET /api/me/features`: Fetches user active plan and feature flags (respects 2-way admin overrides).

- *Response 200*: `{"plan": "premium", "max_medications": 999, "scanner": true, "caregiver_shield": true}`

2.2 Medications & Refills

• `POST /api/medications`: Create medication record.

- *Payload*: `{"name": "Metformin", "dosage": "500", "unit": "mg", "times": ["08:00"], "stock": 30}`

• `POST /api/medications/{id}/refill`: Refill pill inventory.

- *Payload*: `{"add_stock": 30}`

- *Response 200*: `{"ok": true, "stock": 60}`

2.3 Flash Announcements

• `GET /api/announcements/active?plan=free`: Fetches active Swiggy/Zomato style banners.

• `POST /admin/announcements` *(Admin Only)*: Publish flash banner.

• `DELETE /admin/announcements/{id}` *(Admin Only)*: Revoke live announcement.

2.4 Healthcare & Financial Analytics *(Admin Only)*

• `GET /admin/analytics/financials`: Returns MRR, ARR, ARPU, LTV, Churn, Tier Breakdown.

• `GET /admin/analytics/healthcare-intelligence`: Returns adherence rates, refill GMV, risk heatmaps, caregiver dispatch counts.

• `POST /admin/users/{user_id}/features`: 1-Click manual feature override.

Document: Integration_Documentation.md

PillCare - Integration & Third-Party Services Documentation

1. Integrations Summary

Provider	Purpose	Protocol	Fallback / SLA
WhatsApp Business API	Caregiver Missed Dose & Refill Alerts	Webhook / REST POST	Fallback to SMS via Twilio / Gupshup
Google Sign-In OAuth 2.0	Single Sign-On (SSO) Patient Auth	OAuth2 OpenID Connect	Local Device PIN / Biometrics
Android Health Connect / Apple HealthKit	Health Metrics & Vitals Aggregation	Native Device Bridge API	Manual Self-Reported Log Entry
Firebase Cloud Messaging (FCM)	Remote Push Notifications & Banners	HTTP/v1 Protocol	Local Native OS Alarm Manager
Cloudflare DNS & Transform Rules	Subdomain Routing (`admin.pillcare.in`)	Edge Proxy URL Rewrite	Direct Backend Host IP Routing

2. WhatsApp Caregiver Payload Example

```
{
  "to": "+919876543210",
  "type": "template",
  "template": {
    "name": "pillcare_missed_dose_alert",
    "language": { "code": "en_US" },
    "components": [
      {
        "type": "body",
        "parameters": [
          { "type": "text", "text": "Metformin HCl 500mg" },
          { "type": "text", "text": "08:00 AM" }
        ]
      }
    ]
  }
}
```

Module: 04 Security Privacy

Document: Privacy_Policy.md

PillCare - Privacy Policy & Data Sovereignty

Last Updated: 15 September 2026

1. Our Commitment to Patient Privacy

PillCare is committed to protecting patient health information. We adhere strictly to HIPAA (Health Insurance Portability and Accountability Act), DISHA (Digital Information Security in Healthcare Act), and GDPR guidelines.

2. Core Privacy Guarantees

1. **Zero Sale of Health Data**: We NEVER sell, rent, monetise, or commercialize patient health metrics or medication schedules to third parties or advertisers.

2. **Local Storage Sovereignty**: Medication schedules and logs remain encrypted on your device via SQLite (`AsyncStorage`).

3. **Caregiver Escalation Controls**: Caregiver notifications are sent ONLY to contacts explicitly authorized by the user.

3. Data Collected & Purpose

- **Medication Names & Timings**: For firing local alarms and calculating adherence.
- **Caregiver Contact Information**: For emergency WhatsApp/SMS missed-dose escalation.
- **Aggregated Analytics**: Anonymous, de-identified adherence statistics for system performance tuning.

4. User Rights (GDPR & DISHA)

Users have the absolute right to:

- Export their complete health logs in CSV/PDF format.
- Instantly purge their entire history from local device and cloud servers ("Right to be Forgotten").

Document: Terms_Conditions.md

PillCare - Terms & Conditions

Effective Date: 15 September 2026

1. Medical Disclaimer (Non-Diagnostic Advisory Service)

PillCare is an assistive medication reminder and health tracking utility. **PillCare is NOT a licensed medical provider and does NOT provide medical advice, diagnosis, or treatment recommendations.** Always consult a qualified physician or pharmacist before starting, stopping, or altering any medication schedule.

2. Emergency Situations

PillCare is NOT an emergency response system. In case of a medical emergency or severe drug overdose, immediately contact your local emergency services (e.g. 112 / 108) or visit the nearest hospital emergency room.

3. Subscription & Billing Terms

- **Free Plan**: Free access to core reminders for up to 3 active medications.
- **Moksha Pro Subscription (■99/month)**: Unlocks unlimited medications, unlimited caregiver contacts, OCR scanning, and premium wellness yoga flows.
- **Cancellation**: Users may cancel subscriptions at any time via Play Store / App Store subscription settings.

Document: Security_Architecture.md

PillCare - Security Architecture Specification

1. Zero-Trust Security Pillars

```
+-----------------------------------------------------------------------+
|  1. LOCAL DATA ENCRYPTION AT REST (AES-256 via expo-secure-store)     |
+-----------------------------------------------------------------------+
|  2. DATA IN TRANSIT PROTECTION (TLS 1.3 Encryption for all HTTPS REST) |
+-----------------------------------------------------------------------+
|  3. ORM PARAMETERIZATION (Protection against SQLi & XSS Fuzzing)      |
+-----------------------------------------------------------------------+
|  4. ADMIN AUTHENTICATION GATEWAY (HMAC-SHA256 & Header Verification) |
+-----------------------------------------------------------------------+
```

2. Threat Matrix & Defensive Controls

Threat Vector	Severity	Mitigation Control	Verification Result
SQL Injection (SQLi)	**Critical**	Parameterized queries via FastAPI ORM. Malicious inputs like `' DROP TABLE medications; --` handled safely as literal strings.	■ PASSED
Cross-Site Scripting (XSS)	**High**	HTML tag stripping and React Native text node escaping (`<script>` tags neutralized).	■ PASSED
Unauthorized Admin Access	**Critical**	401 Unauthorized block on `/admin/*` routes when `x-admin-secret` or session cookie is missing.	■ PASSED
Man-in-the-Middle (MitM)	**High**	SSL Certificate pinning and strict TLS 1.3 protocol enforcement.	■ PASSED

Document: Data_Retention.md

PillCare - Data Retention & Purge Specification

1. Data Lifecycle Schedule

Data Category	Local Retention	Cloud Retention	Automated Action
Active Medication Schedules	Indefinite (until deleted by user)	Active while account exists	Synchronized upon changes
Dose Confirmation Logs	365 Days	730 Days (Aggregated)	Pruned after 2 years
Caregiver Alert Dispatch Logs	90 Days	180 Days	Soft deleted after 90 days
Swiggy/Zomato Announcement CTR	N/A	30 Days	Aggregated into conversion metrics
Deleted Accounts	Immediate local wipe	30 Days (Purge window)	Permanent hard delete

2. Hard Delete Purge Protocol

Upon user account deletion request:

1. Local SQLite tables (`medications`, `medication_logs`, `caregivers`) dropped/cleared locally.
2. Cloud database user record marked `is_deleted = TRUE` and queued for background purge task after 30 days.

Document: Consent_Management.md

PillCare - Consent Management Matrix

1. User Consent Options & Default States

Consent Scope	Description	Default State	Opt-Out Impact
Caregiver Escalation	Permission to dispatch missed-dose notifications to caregiver WhatsApp/SMS.	**OPT-IN (Required)**	Caregiver Shield notifications disabled.
Health Telemetry	Sync adherence statistics to cloud for Healthcare Intelligence Radar.	**OPT-IN (Default)**	Adherence trends visible only on local device.
Flash Announcements	Display promotional banners (e.g. Swiggy/Zomato partner offers).	**OPT-IN (Default)**	Banner area hidden on Today tab.
Health Connect Integration	Sync vitals (Blood pressure, glucose) from native HealthKit/Health Connect.	**OPT-OUT (Explicit Grant)**	Vitals must be logged manually.

Module: 05 Development

Document: Development_Setup.md

PillCare - Developer Environment Setup Guide

1. System Requirements & Prerequisites

- **Operating System**: macOS 14+ (ARM64 Apple Silicon or x86_64) or Linux (Ubuntu 22.04+).
- **Node.js**: v22.x LTS (with npm 10+ or Yarn 1.22+).
- **Python**: v3.14 (or v3.11+).
- **Java JDK**: OpenJDK 17 (`/opt/homebrew/opt/openjdk@17`).
- **Android SDK**: API Level 36, Build-Tools 35.0.0 / 36.0.0, NDK 27.1 (`r27b`).

2. Setting Up the Backend

```
cd backend
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
python server.py
```

Backend will start on `http://localhost:8000`.

3. Setting Up the Frontend

```
cd frontend
npm install
npx expo start
```

4. Local Android Build Execution

To compile a local debug/release APK directly:

```
cd frontend/android
JAVA_HOME=/opt/homebrew/opt/openjdk@17/libexec/openjdk.jdk/Contents/Home \
ANDROID_HOME=~/Library/Android/sdk \
./gradlew assembleDebug
```

Output APK generated at `frontend/android/app/build/outputs/apk/debug/app-debug.apk`.

Document: Coding_Standards.md

PillCare - Coding Standards & Engineering Guidelines

1. TypeScript & React Native Standards

- **Strict Mode**: `noImplicitAny`, `strictNullChecks` enforced in `tsconfig.json`.

- **Component Architecture**: Functional components only; use hooks (`useState`, `useEffect`, `useCallback`, `useMemo`).
- **File Naming**: PascalCase for components (`RefillModal.tsx`), camelCase for utilities (`analytics.ts`).
- **Styles**: Use Vanilla CSS / React Native `StyleSheet.create`. No ad-hoc inline styling objects inside render loops.

2. Python Backend Standards (FastAPI)

- **PEP 8 Compliance**: Follow PEP 8 formatting rules.
- **Type Annotations**: All endpoint function signatures must include Python type hints (`Pydantic` schemas).
- **Asynchronous Execution**: Use `async def` for I/O bound endpoints.

3. Strict Directive: Code Push Permission Rule

> [!CAUTION]

> **USER CONSENT MANDATORY**: Per developer directive, no code changes or documentation updates shall be pushed to remote Git (`git push origin main`) without presenting the changes and obtaining explicit user consent.

Document: Git_Workflow.md

PillCare - Git Workflow & Release Strategy

1. Branching Strategy

```
main (Production Deployment & Releases)
 ^
 |  (Pull Request after QA Approval)
feature/*  or  fix/*
```

- `main`: Production ready code. Automatically triggers GitHub Actions build pipelines.
- `feature/<name>`: Isolated feature development branches.
- `fix/<bug-id>`: Bug fixes and hotfixes.

2. Commit Message Conventions

Follow Conventional Commits format:

- `feat(analytics)`: Add Healthcare Intelligence endpoints
- `fix(ci)`: Fix gradlew script path in GitHub Actions workflow
- `docs(prd)`: Update Product Requirements Document
- `refactor(ui)`: Optimize RefillModal render cycle

3. Mandatory User Permission Rule for Git Push

Before running `git push origin main`:

1. Verify `npx tsc --noEmit` returns exit code 0.
2. Run backend 17-point monkey test (`python backend/scratch/full_monkey_end_to_end_test.py`).
3. **STOP AND REQUEST USER CONSENT**: Confirm with the user before executing `git push`.

Document: Environment_Configuration.md

PillCare - Environment Configuration Reference

1. Frontend Environment Variables (`frontend/.env`)

```
# Backend API Base URL
EXPO_PUBLIC_BACKEND_URL=https://pillcare-backend-1082668880575.europe-west1.run.app

# Google OAuth Client IDs
EXPO_PUBLIC_GOOGLE_WEB_CLIENT_ID=1082668880575-lfoqgnieodb52b5ges1vppiv92ifbioc.apps.googleusercontent.com
EXPO_PUBLIC_GOOGLE_IOS_CLIENT_ID=1082668880575-og9vm3sduk1g7dlj8g17in3r1cicnbt6.apps.googleusercontent.com
EXPO_PUBLIC_GOOGLE_ANDROID_CLIENT_ID=1082668880575-5hrg3n2acflsjtf8bo9ih7b2dsecb9j2.apps.googleusercontent.com
```

2. Backend Environment Variables (`backend/.env`)

```
PORT=8000
ADMIN_SECRET_KEY=pillcare_admin_secret_key_2026
DATABASE_URL=sqlite:///pillcare.db
WHATSAPP_API_TOKEN=your_whatsapp_token_here
```

Module: 06 QA

Document: Test_Plan.md

PillCare - QA Master Test Plan

1. Test Strategy Overview

The QA strategy for PillCare combines automated end-to-end integration tests, security fuzz testing (SQLi/XSS), TypeScript strict static analysis, and manual senior-focused UAT.

2. Testing Levels & Tools

Level	Tool / Framework	Target Scope	Execution Frequency
Static Analysis	TypeScript (`npx tsc --noEmit`)	Type safety & API contracts	Pre-commit / CI
Backend Integration	Python `httpx` / `pytest`	Async API endpoints & Database ORM	Daily / Automated
Fuzz & Vulnerability	Python Fuzzing Scripts	SQLi & XSS injection resilience	Pre-release
End-to-End Monkey	Custom Monkey Test Runner	17-Point complete system validation	Pre-release
Mobile Native UI	Expo Go / Android Emulator	Alarm triggers & UI touch targets	Release Candidate

Document: Test_Cases.md

PillCare - QA Test Cases & Automation Suite

1. Automated Test Suite Execution Results (17/17 PASSED)

```
================================================================================
      PILLCARE FULL MONKEY TEST & ADMIN-TO-APP DATA WIRING VERIFICATION
================================================================================

--- 1. SECURITY & ADMIN AUTHENTICATION ---
  ■ [PASS] Unauthorized /admin/stats blocked with 401
  ■ [PASS] Valid Admin Secret access granted to /admin/stats

--- 2. ADMIN-TO-APP 2-WAY MONETIZATION WIRING ---
  ■ [PASS] User initially registered on Free Plan default (Max Meds: 3)
  ■ [PASS] Admin successfully upgraded user to Premium Tier
  ■ [PASS] Mobile App immediately received Premium Plan features (OCR Scanner: Enabled, Max Meds: 999)
  ■ [PASS] Admin set manual feature override 'scanner=False' on Premium user
  ■ [PASS] Mobile App correctly respects Admin 1-Click Feature Override precedence (Scanner: Forced Disabled
)

--- 3. SWIGGY / ZOMATO FLASH ANNOUNCEMENT WIRING ---
  ■ [PASS] Admin published live Flash Announcement
```

```
  ■ [PASS] Mobile App received real-time Swiggy/Zomato Flash Announcement on Today Screen banner!
  ■ [PASS] Admin revoked Flash Announcement

--- 4. HEALTHCARE INTELLIGENCE & FINANCIAL ANALYTICS ---
  ■ [PASS] JP Morgan Financial Analytics: MRR=■990, ARR=■11,880, ARPU=■110.0
  ■ [PASS] Healthcare Intelligence Analytics: Refill GMV=■2,450, Caregiver Alerts Active

--- 5. CORE PATIENT EXPERIENCE & FUZZ SAFETY ---
  ■ [PASS] Created medication 'Metformin HCl'
  ■ [PASS] Malicious SQLi/XSS payload handled safely by backend ORM
  ■ [PASS] Medication Refill stock updated to 40
  ■ [PASS] AI Yoga Therapy Session logged to progress journal
  ■ [PASS] Caregiver Alert escalation logged and dispatched to caregiver

======================================================================
     MONKEY TEST COMPLETE: 17 PASSED, 0 FAILED out of 17 TESTS
======================================================================
```

Document: UAT.md

PillCare - User Acceptance Testing (UAT) Checklist

1. Senior Patient Usability Validation

- [x] **Large Text Visibility**: Text sizes legible without reading glasses ($\ge 18\text{sp}$).
- [x] **Touch Target Precision**: Buttons easily tappable ($\ge 48\text{dp}$ touch bounding box).
- [x] **Audio Alarm Clarity**: High-pitch wellness audio alerts audible across room noise.
- [x] **1-Tap Take Action**: Single tap marks medication as taken with instant haptic confirmation.

2. Offline Resilience Validation

- [x] **Airplane Mode Verification**: Turn on Airplane mode $\rightarrow$ Alarms fire on time $\rightarrow$ Dose logged locally $\rightarrow$ Syncs automatically upon network reconnection.

Document: Bug_Reports.md

PillCare - QA Bug Reports & Historical RCA Log

1. Historical Resolution Log

BUG-001: GitHub Actions CI `chmod` Directory Path Failure

- **Severity**: Critical (CI/CD Pipeline Failure)
- **Root Cause**: `working-directory: frontend` evaluated `chmod +x android/gradlew` relative to `frontend`, causing `chmod: cannot access 'android/gradlew': No such file or directory`.
- **Resolution**: Updated script to `cd frontend/android && chmod +x gradlew && ./gradlew assembleRelease`.

BUG-002: GitHub Actions C++ Compilation Memory Exhaustion

- **Severity**: High (CI Runner Hang >40 mins)
- **Root Cause**: Compiling C++ TurboModules for 4 ABIs in parallel exhausted memory on 7GB `ubuntu-latest` runners.
- **Resolution**: Added `-PreactNativeArchitectures=arm64-v8a` and `--max-workers=2` to constrain memory usage.

BUG-003: EAS CLI Project Permission Error

- **Severity**: Medium (Cloud Build Submission Error)
- **Root Cause**: Account handle mismatch between CLI session (`pillcare`) and browser org (`pillcare-reminder`).
- **Resolution**: Updated `app.json` owner to `pillcare-reminder` and authenticated via `EXPO_TOKEN`.

Module: 07 Deployment

Document: Deployment_Guide.md

PillCare - Production Deployment Guide

1. Cloud Infrastructure Overview

- **Backend Service**: GCP Cloud Run (Serverless Python 3.14 container scaling `0 -> 100` instances).
- **Domain Edge & DNS**: Cloudflare CDN (`admin.pillcare.in` routed to `/admin.html`).
- **Mobile Distribution**: Expo EAS Cloud & Google Play Console (Internal/Production Tracks).

2. GCP Cloud Run Deployment Execution

```
# Build and deploy backend container to GCP Cloud Run
gcloud builds submit --tag gcr.io/pillcare-prod/backend:latest backend/
gcloud run deploy pillcare-backend \
  --image gcr.io/pillcare-prod/backend:latest \
  --platform managed \
  --region europe-west1 \
  --allow-unauthenticated
```

3. Cloudflare Routing Rules Configuration

1. **CNAME Record**: `admin.pillcare.in` $\rightarrow$ `pillcare.in` (Proxied `ORANGE CLOUD`).

2. **Transform URL Rewrite Rule**:

- Expression: `(http.host eq "admin.pillcare.in" and http.request.uri.path eq "/")`

- Rewrite: Dynamic Path `/admin.html`

Document: CI_CD.md

PillCare - CI/CD Pipeline Documentation

1. Pipeline Overview (`.github/workflows/build-android.yml`)

```
  Push to main Branch
         |
         v
+-------------------------------------------------------------+
| GitHub Actions Runner (ubuntu-latest)                        |
|  1. Checkout code & Setup JDK 17 / Node.js 22               |
|  2. npm ci (Install dependencies)                           |
|  3. npx expo prebuild --platform android --clean             |
|  4. ./gradlew assembleRelease -PreactNativeArchitectures=... |
|  5. Upload app-release.apk to GitHub Action Artifacts       |
+-------------------------------------------------------------+
```

2. Expo EAS Cloud Pipeline

Triggered via `EXPO_TOKEN` authenticated CLI:

```
EXPO_TOKEN=<token> npx eas build --platform android --profile preview --non-interactive
```

Generates cloud build link and downloadable standalone APK.

Document: Release_Checklist.md

PillCare - Pre-Release Checklist

1. Quality & Security Checks

- [x] **TypeScript Validation**: `npx tsc --noEmit` returns 0 errors.
- [x] **Monkey End-to-End Test**: `python backend/scratch/full_monkey_end_to_end_test.py` passes 17/17 tests.
- [x] **SQLi & XSS Fuzz Audit**: Parameterized ORM sanitization verified.
- [x] **Admin Authorization Lock**: `/admin/*` routes verify `x-admin-secret` header.

2. Environment & Build Checks

- [x] `EXPO_PUBLIC_BACKEND_URL` pointing to production GCP Cloud Run endpoint.
- [x] `app.json` versionCode incremented (`versionCode = 1`).
- [x] Android Release APK compiled (`app-release.apk` 112 MB verified).
- [x] **User Consent Verification**: User explicit permission requested before git push.

Document: Rollback_Procedure.md

PillCare - Emergency Rollback Procedure

1. Backend Service Rollback (GCP Cloud Run)

If a critical regression is detected in production:

```
# List previous revisions
gcloud run revisions list --service pillcare-backend --region europe-west1

# Route 100% traffic back to previous stable revision
gcloud run services update-traffic pillcare-backend \
  --to-revisions pillcare-backend-00042-xyz=100 \
  --region europe-west1
```

2. Play Store Mobile App Rollback Strategy

- Android native apps cannot roll back binary versions directly on Play Store.
- **Hotfix Strategy**: Revert git commit locally, increment `versionCode`, recompile APK/AAB, and publish emergency update track within 2 hours.

Module: 08 Admin Panel

Document: Requirements.md

PillCare Admin Panel - Mission Control Requirements

1. Overview & Capabilities

The PillCare Admin Panel (`admin.html` hosted live on `admin.pillcare.in`) provides real-time operational management, financial monitoring, healthcare intelligence analytics, and 2-way client controls.

2. Core Functional Requirements

2.1 Security & Auth Gateway

- Google Sign-In Single Sign-On (SSO) with DOM fallback.
- Mandatory `x-admin-secret` header verification (`pillcare_admin_secret_key_2026`).

2.2 JP Morgan Grade Financial Dashboard

- **MRR (Monthly Recurring Revenue)**: Calculated from active Pro subscriptions.
- **ARR (Annual Run Rate)**: $\text{MRR} \times 12$.
- **ARPU (Average Revenue Per User)**: $\text{Total Revenue} / \text{Active Users}$.
- **LTV (Lifetime Value)** & **Churn Rate**: Calculated across subscription cohorts.

2.3 Healthcare Intelligence & Refill Radar

- **Adherence Rate Tracker**: Percentage of scheduled doses confirmed taken vs missed.
- **Estimated Refill GMV (Gross Merchandise Value)**: Projected pharmacy prescription value.
- **Caregiver Alert Counters**: Total missed-dose alerts dispatched.
- **Condition Distribution Heatmap**: Patient distribution across Hypertension, Diabetes, Asthma, etc.

2.4 Swiggy / Zomato Flash Announcement Publisher

- Form to create title, body, plan targeting (`all`, `free`, `pro`), type (`offer`, `alert`), and link URL.
- 1-Click announcement revocation button.

2.5 2-Way Manual Feature Overrides

- Search user by ID or email.
- Override subscription tier (Free $\rightarrow$ Pro/Enterprise).
- Toggle specific feature flags (`scanner`, `caregiver_shield`, `export`, `max_medications`).

Document: RBAC.md

PillCare Admin Panel - Role-Based Access Control (RBAC)

1. RBAC Matrix

Role	Financial KPIs	Healthcare Intel	Flash Announcements	Feature Overrides	System Audit Logs
Super Admin	Full Access	Full Access	Publish / Delete	Full Overrides	View & Export
Support Specialist	Read Only	Patient Summary	View Only	Modify User Plan	No Access
Marketing Manager	Revenue CTR	Conversion Data	Publish / Delete	No Access	View Banners
Medical Auditor	No Access	Anonymized Intel	No Access	No Access	View Compliance

2. API Scope Matrix

- `x-admin-role: superadmin` required for manual feature overrides (`POST /admin/users/{id}/features`).
- `x-admin-role: marketing` required for announcement publishing (`POST /admin/announcements`).

Document: Admin_Functions.md

PillCare Admin Panel - Operational Procedures & Manual Controls

1. Upgrade User Plan Manually

1. Log into `admin.pillcare.in`.
2. Scroll to **2-Way User Feature Overrides** table.
3. Locate Target User ID.
4. Select desired plan (`pro` / `enterprise`) and duration (`30 days`).
5. Click **Apply Upgrade**. Backend immediately updates user record; mobile app reflects new plan limits on next feature sync (`/api/me/features`).

2. Publish Swiggy/Zomato Flash Offer

1. Open **Flash Announcement Publisher** section.
2. Fill Title: `■ Special 50% Off Annual Moksha Pro`.
3. Select Target Audience: `Free Users`.
4. Enter Link URL: `https://pillcare.in/upgrade`.
5. Click **Publish Announcement Live**. Banner immediately appears at the top of the Today screen for all Free plan mobile users.

Document: Audit_Logs.md

PillCare Admin Panel - Audit Logs & Compliance Security

1. Audit Log Capture Schema

All administrative actions (plan upgrades, feature overrides, announcement publishing, revenue queries) are automatically recorded in the audit log database table:


```
{
  "id": "audit_91823102",
  "timestamp_iso": "2026-09-15T00:45:00Z",
  "admin_user": "nitinrai264@gmail.com",
  "action": "USER_PLAN_UPGRADE",
  "target_user_id": "test_user_676cf52c",
  "details": {
    "previous_plan": "free",
    "new_plan": "premium",
    "override_note": "Hospital Enterprise Contract"
  },
  "ip_address": "103.21.124.5"
}
```


2. Security Alerts

- **Repeated Auth Failure Alert**: Triggered if 5+ invalid `x-admin-secret` headers are received within 60 seconds.

Module: 09 Play Store

Document: Store_Listing.md

Google Play Store Listing Copy - PillCare Reminder

1. Store Metadata

- **App Name**: PillCare: Smart Pill Reminder
- **Short Description (80 characters)**: Smart offline pill reminder, refill radar & caregiver safety shield for family.
- **Category**: Medical / Health & Fitness
- **Content Rating**: Everyone

2. Full Description (4000 characters)

PillCare is India's most trusted offline-first medication reminder, refill automation, and caregiver emergency shield app. Built for senior citizens, patients managing chronic conditions, and busy families, PillCare ensures you never miss a dose.

■ KEY FEATURES:

■ OFFLINE-FIRST EXACT ALARMS

- Works 100% offline without active internet.
- High-volume alarm sound options and custom voice alerts.
- Bypasses battery optimization so alarms fire on time, every time.
- Flexible schedules: Daily, alternate days, interval cycles, PRN (As Needed).

■■ CAREGIVER SHIELD (EMERGENCY ALERTS)

- Automatically alerts family members or caregivers via WhatsApp if a medication is missed for more than 15 minutes.
- Gives loved ones peace of mind no matter where they live.

■ REFILL RADAR & AUTOMATED INVENTORY

- Automatically tracks remaining pill counts.
- Low-stock alert notifies you before your pills run out.
- 1-Tap reordering via your preferred pharmacy.

■ AI PRESCRIPTION OCR SCANNER

- Snap a photo of any doctor's prescription.
- Automatically recognizes drug names, dosages, and daily schedules.

■ WELLNESS & YOGA THERAPY

- Guided gentle pranayama and condition-specific yoga flows for hypertension, diabetes, and joint flexibility.
- Daily health habit journal and streak rewards.

■ MEDICAL PRIVACY GUARANTEE

- We NEVER sell your medical data.
- AES-256 encrypted local database storage.
- Full HIPAA & DISHA compliant security architecture.

Download **PillCare** today and protect your health and your family's peace of mind!

Document: Data_Safety.md

Google Play Console Data Safety Form Declarations

1. Declarations Summary

Question	Answer	Notes
Does your app collect or share user data?	**YES**	Medication names & caregiver contact phone numbers for alert escalation.
Is all user data encrypted in transit?	**YES**	TLS 1.3 HTTPS protocol strictly enforced.
Do you provide a way for users to request data deletion?	**YES**	In-app 1-tap delete account & local storage wipe.
Is data collection optional?	**NO**	Core functionality (alarm scheduling) requires local storage of med names.

2. Data Types Declared

1. **Health Info (Health & Fitness)**:

- Data Type: Medications, dosage schedules, adherence percentage.

- Purpose: App functionality, analytics.

- Encryption: Encrypted in transit & at rest.

2. **Personal Info (Name, Email, Phone Number)**:

- Data Type: User name & Caregiver phone number.

- Purpose: Emergency alert dispatch via WhatsApp API.

- Sharing: Shared ONLY with authorized caregiver number specified by patient.

Document: Release_Documentation.md

Google Play Store Release Documentation

1. Versioning Specification

- `version`: `1.0.0`
- `versionCode`: `1` (Increments +1 per Play Console release).
- **Target SDK**: Android 16 (API 36).
- **Min SDK**: Android 8.0 (API 26).

2. Release Artifact Generation Command

To build the signed Android App Bundle (`.aab`) for Google Play Store upload:

```
cd frontend/android
JAVA_HOME=/opt/homebrew/opt/openjdk@17/libexec/openjdk.jdk/Contents/Home \
ANDROID_HOME=~/Library/Android/sdk \
./gradlew bundleRelease -PreactNativeArchitectures=arm64-v8a
```

Output bundle generated at `frontend/android/app/build/outputs/bundle/release/app-release.aab`.

Module: 10 Operations

Document: SOPs.md

PillCare Operations - Standard Operating Procedures (SOPs)

1. SOP-001: Daily Backend Health Check

1. Execute health check HTTP GET against GCP Cloud Run container:

`bash

curl -i https://pillcare-backend-1082668880575.europe-west1.run.app/health

`

2. Verify HTTP status is `200 OK`.

3. Check Cloud Run latency graphs on GCP Console ($<100\text{ms}$ median).

2. SOP-002: Provisioning Hospital Enterprise Subscriptions

1. Log into `admin.pillcare.in`.

2. Locate hospital patient list or input user ID.

3. Apply plan upgrade: `enterprise` with 365-day validity.

4. Verify 2-way override is stored in database.

Document: Incident_Management.md

PillCare Incident Management & Response Framework

1. Incident Severity Definitions

Severity	Definition	Target Response (TTR)	Target Resolution (RTO)	Escalation Contact
SEV-1 (Critical)	Core notification engine down, backend DB unreachable, or active security breach affecting >10% of active users.	<15 minutes	<1 hour	VP Engineering + On-Call Lead
SEV-2 (High)	Caregiver sync failure, Firestore sync delays >5 mins, or partial API failure impacting non-critical screens (e.g. Refill UI).	<30 minutes	<4 hours	Lead Backend Engineer

SEV-3 (Medium)	Non-blocking UI glitch, slow analytics logging, or single region API latency spikes ($>500\text{ms}$).	<2 hours	<24 hours	On-Call Engineer
SEV-4 (Low)	Minor cosmetic issue, documentation typo, or non-user-impacting administrative tool bug.	Next Business Day	Next Sprint	Team Backlog

2. Incident Response Workflow

```
graph TD
    A[Alert Triggered / User Report] --> B{Determine Severity}
    B -->|SEV-1 / SEV-2| C[Declare Incident & Create Slack #incident Channel]
    B -->|SEV-3 / SEV-4| D[Log Jira Ticket & Assign to Backlog]
    C --> E[Assign Incident Commander]
    E --> F[Investigate Root Cause & Mitigate]
    F --> G{Mitigated?}
    G -->|No| H[Escalate to Executive Team & Apply Workaround]
    G -->|Yes| I[Verify System Health & Monitor]
    I --> J[Close Incident & Schedule PIR]
```

Steps:

1. **Identification**: Automated GCP Cloud Monitoring alarm or user incident ticket.
2. **Containment**: Immediately isolate affected Cloud Run instances or roll back to last known stable container tag.
3. **Communication**: Update status page (`status.pillcare.in`) within 15 minutes for SEV-1/SEV-2.
4. **Resolution**: Deploy hotfix via CI/CD pipeline or apply emergency configuration change via GCP Secret Manager.
5. **Post-Incident Review (PIR)**: Conduct blameless root cause analysis within 48 hours for all SEV-1 and SEV-2 events.

3. On-Call Runbooks

Runbook A: Alarm Triggered for Notification Failures

1. Check Cloud Watch / Firebase Cloud Messaging (FCM) error status codes.
2. Verify FCM server key validity in GCP Secret Manager (`FCM_SERVER_KEY`).
3. Test payload dispatch via `curl`:

`bash
curl -X POST -H "Authorization: key=$FCM_SERVER_KEY" \
-H "Content-Type: application/json" \
-d '{"to":"TEST_DEVICE_TOKEN","notification":{"title":"Test","body":"Ping"}}' \
https://fcm.googleapis.com/fcm/send
`

4. If FCM API returns 401/403, rotate key via Secret Manager and deploy hotfix.

Runbook B: High Database Connections / Thread Exhaustion

1. Check Postgres active connection count:

`sql

SELECT count(*) FROM pg_stat_activity WHERE state = 'active';
`

2. Terminate long-running locking queries if necessary:

`sql
SELECT pg_terminate_backend(pid) FROM pg_stat_activity WHERE duration > interval '5 minutes';
`

3. Scale Cloud Run max instances to handle spike while connection pooler resets.

Document: Backup_Recovery.md

PillCare Disaster Recovery & Backup Strategy

1. Recovery Objectives

- **Recovery Point Objective (RPO)**: $<15\text{ minutes}$ (Maximum acceptable data loss window)
- **Recovery Time Objective (RTO)**: $<1\text{ hour}$ (Maximum acceptable downtime window)

2. Backup Schedules & Retentions

Resource	Backup Method	Frequency	Retention Period	Storage Location
Cloud SQL (PostgreSQL)	Automated GCP Snapshots	Daily at 02:00 UTC	30 Days	GCP Multi-Region (`eu`)
Cloud SQL Write Logs	Point-In-Time Recovery (PITR) WAL	Continuous	7 Days	GCP Storage Bucket (`gs://pillcare-pitr-wal`)
Firestore NoSQL State	Export Command via Cloud Scheduler	Every 6 hours	14 Days	GCS Nearline Storage
App Configuration & Secrets	Terraform State + Secret Manager	Version Controlled / Commit Tag	Infinite	Private GitHub + GCP KMS
User Media & Attachments	GCS Bucket Replication	Real-time Object Versioning	90 Days	GCS Dual-Region (`europe-west1` + `europe-west4`)

3. Disaster Recovery Procedures

Scenario A: Database Corruption or Loss (Point-in-Time Restore)

1. Identify exact time of corruption ($T_{corrupt}$).

2. Execute Cloud SQL PITR command:

`bash
gcloud sql instances restore-instance pillcare-postgres-db \
--restore-instance=pillcare-postgres-db-pitr-target \
--backup-instance=pillcare-postgres-db \

--point-in-time="2026-09-15T23:45:00Z"
`

3. Verify table integrity using database validation scripts.
4. Update Cloud Run DB connection string environment variable to target restored instance.
5. Re-enable traffic via Cloud Run revisions.

Scenario B: Regional GCP Outage

1. Initiate failover to secondary GCP region (`europe-west4`):

`bash
gcloud run deploy pillcare-backend \
--region=europe-west4 \
--image=europe-west1-docker.pkg.dev/pillcare-prod/app/backend:latest
`

2. Update Cloud DNS A/AAAA records to point `api.pillcare.in` to secondary region load balancer IP.
3. Validate client app latency and notification delivery status.

---

4. Disaster Recovery Testing Schedule

- **Tabletop Exercises**: Conducted quarterly with engineering leads.
- **Failover Drills**: Conducted semi-annually in staging environment.
- **Backup Restoration Validation**: Automated monthly test restore pipeline to isolated sandbox DB.

Document: Maintenance.md

PillCare System Maintenance & Patch Management

1. Maintenance Windows

- **Routine Operations**: Scheduled every Tuesday between 02:00 UTC and 04:00 UTC.
- **Zero-Downtime Policy**: All backend API updates must be deployed using Cloud Run traffic splitting / blue-green deployments. Database schema changes must follow non-breaking multi-phase migrations.
- **Maintenance Notification Window**: 72 hours advance notice required for any disruptive maintenance via in-app banner and push notification.

---

2. Maintenance Tasks Checklist

Weekly Tasks

- [ ] Review GCP Cloud Monitoring error rate graphs ($<0.01\%$ threshold).
- [ ] Inspect automated dependency vulnerability scans (Dependabot / Snyk).
- [ ] Clean up transient storage artifacts in GCS scratch buckets.

Monthly Tasks

- [ ] Apply OS and security patches to backend base Docker images (`node:20-alpine` updates).

- [ ] Review IAM permission grants in GCP & Firebase Console; revoke stale access.
- [ ] Conduct automated database index fragmentation check and re-index if needed.
- [ ] Verify SSL/TLS certificate auto-renewals on Cloud Load Balancer.

Quarterly Tasks

- [ ] Execute full Disaster Recovery restoration drill.
- [ ] Review API rate limiting thresholds and DDOS protection rules on Cloud Armor.
- [ ] Perform quarterly audit log review with compliance officer.

3. Planned Maintenance Procedure

1. **Pre-Maintenance**:

- Create staging snapshot of database.

- Publish in-app announcement banner via Admin Panel.

2. **Execution**:

- Execute database migration scripts (`npx prisma migrate deploy` or equivalent).

- Deploy updated Cloud Run backend service container.

- Route 10% traffic to new revision; monitor logs for 15 minutes.

- Increase traffic to 100% if error rates remain $<0.05\%$.

3. **Post-Maintenance**:

- Verify health check endpoints (`GET /health`).

- Remove maintenance banner.

- Document completion in Maintenance Log.